

Paul D. Franzon

professor

基本信息

姓名: Paul D. Franzon

所属院校: 北卡罗来纳州立大学-罗利

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荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

研究领域

低功耗电路与能量管理

数字集成电路设计

EDA工具与电路验证

通信系统仿真与优化

低功耗设计与能耗优化

深度学习与神经网络

计算机体系结构

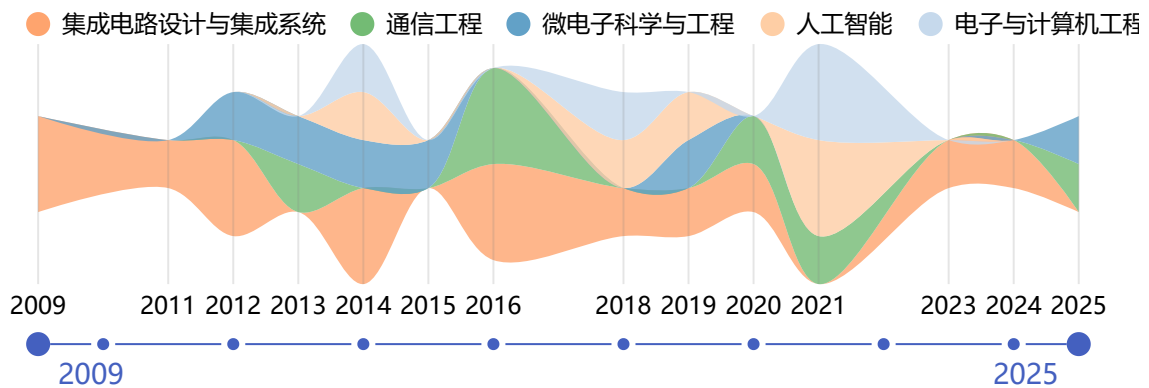
超大规模集成电路设计

个人简介

Paul D. Franzon is a professor of Electrical and Computer Engineering at NCSU. His teaching and research focus on microsystems and nanosystems, covering areas like 3D processor design, interconnect structures, cognitive computing, application - specific chips, microsystems, and emerging devices. He is the director of the Microelectronics Systems Laboratory and co - director of the Electronics Research Laboratory. He also co - directs NCSU's CAD efforts. A fellow of the IEEE, he has chaired conferences, served on committees, and

been an associate editor for journals. Contact him at paulf@ncsu.edu. His personal interests include flying, horse riding, and more, and he has a military background.

近年发表的论文



A 27-30 GHz T/R Module With Reflection-Type Phase Shifting and Machine-Learned Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chang, Yuan; Wen, Yuejiang; Hong, Zhangjie; Padmanabhan, Bharadwaj; Franzon, Paul D.; Floyd, Brian A.

被引用次数: 0

2025

Can Low-Rank Knowledge Distillation in LLMs be Useful for Microelectronic Reasoning?

2024 IEEE LLM AIDED DESIGN WORKSHOP, LAD 2024

Rouf, Nirjhor; Amin, Fin; Franzon, Paul D.

被引用次数: 0

2024

System Aware Floorplanning for Chip-Package Co-design

2023 IEEE 32ND CONFERENCE ON ELECTRICAL PERFORMANCE OF ELECTRONIC PACKAGING AND SYSTEMS, EPEPS

Pan, Tse-Han; Franzon, Paul D.; Srinivas, Vaishnav; Nagarajan, Mahalingam; Popovic, Darko

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2023

FINITE ELEMENT ANALYSIS AND FATIGUE LIFE PREDICTION OF A Laterally Conducting GAN-BASED POWER PACKAGE UNDER THERMAL CYCLING

PROCEEDINGS OF ASME 2023 INTERNATIONAL MECHANICAL ENGINEERING CONGRESS AND EXPOSITION, IMECE2023, VOL 12

Zaghari, Pouria; Sinha, Sourish S.; Ryu, Jong Eun; Franzon, Paul D.; Hopkins, Douglas C.

被引用次数: 1

2023

A Scalable Cluster-based Hierarchical Hardware Accelerator for a Cortically Inspired Algorithm

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Dey, Sumon; Baker, Lee; Schabel, Joshua; Li, Weifu; Franzon, Paul D.

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2021

High Speed Receiver Modeling Using Generative Adversarial Networks

IEEE 30TH CONFERENCE ON ELECTRICAL PERFORMANCE OF ELECTRONIC PACKAGING AND SYSTEMS (EPEPS 2021)

Kashyap, Priyank; Pitts, W. Shepherd; Baron, Dror; Wong, Chau-Wai; Wu, Tianfu; Franzon, Paul D.

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A Review of 3D-Dynamic Random-Access Memory based Near-Memory Computation

2021 IEEE INTERNATIONAL 3D SYSTEMS INTEGRATION CONFERENCE (3DIC)

Ravichandiran, Prasanth Prabu; Franzon, Paul D.

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2021

Multi-Fidelity Surrogate-Based Optimization for Electromagnetic Simulation Acceleration

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Wang, Yi; Franzon, Paul D.; Smart, David; Swahn, Brian

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2020

Asymmetric Transformer Design With Multiband Frequency Response for Simultaneous Power and Data Transfer

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Zannat, Nazia; Franzon, Paul D.

被引用次数: 1

2020

Appliance Identification Algorithm for a Non-Intrusive Home Energy Monitor Using Cogent Confabulation

IEEE TRANSACTIONS ON SMART GRID

Park, Sung Wook; Baker, Lee B.; Franzon, Paul D.

被引用次数: 17

2019

3-D-DATE: A Circuit-Level Three-Dimensional DRAM Area, Timing, and Energy Model

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Park, Jong Beom; Davis, William Rhett; Franzon, Paul D.

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2019

RF-Only Logic: an Area Efficient Logic Family for RF-Power Harvesting Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhao, Wenxu; Gadfort, Peter; Bhanushali, Kirti; Franzon, Paul D.

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2018

Exploring the Tradeoffs of Application-Specific Processing

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Schabel, Joshua C.; Franzon, Paul D.

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2018

A Generally Applicable Calibration Algorithm for Digitally Reconfigurable Self-Healing RFICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wyers, Eric J.; Morton, Matthew A.; Sollner, T. C. L. Gerhard; Kelley, C. T.; Franzon, Paul D.

被引用次数: 5

2016

Multimode High-Density Link Design Methodology and Implementation

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Yan, Zhuo; Aygun, Kemal; Braunisch, Henning; Franzon, Paul D.

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2016

A Multitier Study on Various Stacking Topologies of TSV-Based PDN Systems Using On-Chip Decoupling Capacitor Models

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Charles, Gary; Franzon, Paul D.

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2015

A Generic and Scalable Architecture for a Large Acoustic Model and Large Vocabulary Speech Recognition Accelerator Using Logic on Memory

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bapat, Ojas A.; Franzon, Paul D.; Fastow, Richard M.

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2014

Thermal Pathfinding for 3-D ICs

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Priyadarshi, Shivam; Davis, W. Rhett; Steer, Michael B.; Franzon, Paul D.

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2014

Adaptive and Reliable Clock Distribution Design for 3-D Integrated Circuits

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Chen, Xi; Zhu, Ting; Davis, William Rhett; Franzon, Paul D.

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2014

Crosstalk-Canceling Multimode Interconnect Using Transmitter Encoding

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, HoonSeok; Won, Chanyoun; Franzon, Paul D.

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2013

A Transient Electrothermal Analysis of Three-Dimensional Integrated Circuits

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Harris, T. Robert; Priyadarshi, Shivam; Melamed, Samson; Ortega, Carlos; Manohar, Rajit; Dooley, Steven R.; Kriplani, Nikhil M.; Davis, W. Rhett; Franzon, Paul D.; Steer, Michael B.

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2012

Comparing Through-Silicon-Via (TSV) Void/Pinhole Defect Self-Test Methods

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Lou, Yi; Yan, Zhuo; Zhang, Fan; Franzon, Paul D.

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2012

Parallel Transient Simulation of Multiphysics Circuits Using Delay-Based Partitioning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Priyadarshi, Shivam; Saunders, Christopher S.; Kriplani, Nikhil M.; Demircioglu, Harun; Davis, W. Rhett; Franzon, Paul D.; Steer, Michael B.

被引用次数: 42012

Accurate and Scalable IO Buffer Macromodel Based on Surrogate Modeling

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Zhu, Ting; Steer, Michael B.; Franzon, Paul D.

被引用次数: 282011

Application Exploration for 3-D Integrated Circuits: TCAM, FIFO, and FFT Case Studies

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Davis, W. Rhett; Oh, Eun Chu; Sule, Ambarish M.; Franzon, Paul D.

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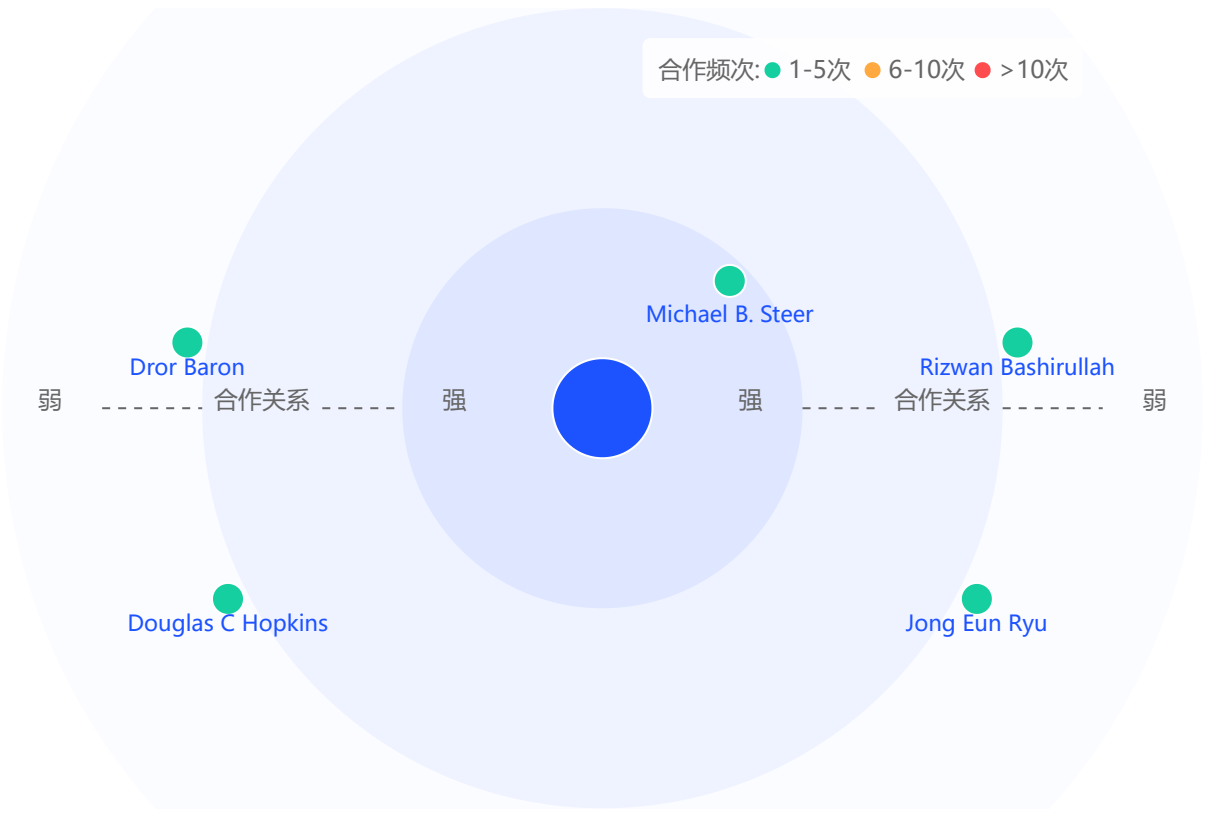
A 32-Gb/s On-Chip Bus With Driver Pre-Emphasis Signaling

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Liang; Wilson, John M.; Bashirullah, Rizwan; Luo, Lei; Xu, Jian; Franzon, Paul D.

被引用次数: 142009

学术合作



Michael B. Steer

合作次数: 4

北卡罗来纳州立大学-罗利

Rizwan Bashirullah

合作次数: 1

佛罗里达大学

Jong Eun Ryu

合作次数: 1

北卡罗来纳州立大学-罗利

Douglas C Hopkins

合作次数: 1

北卡罗来纳州立大学-罗利

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